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DEPARTMENT:-

BS-CYBER SECURITY-FALL-24

SUBJECT:-

PROGRAMMING FUNDAMENTALS (Lab)

PRESENTED BY

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PROVIDED TO

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LAB #05

TASK 1

- **Right Angle Triangle**

Hyp = ?

$$c^2 = a^2 + b^2$$

$$c = \sqrt{a^2 + b^2}$$

Code:

[*] Task 5.1 A.cpp | Lab 5A-Task 1.cpp

```cpp
#include <iostream>
#include <cmath>
using namespace std;
int main ()
{
    //Declaring variables
    float base, hyp, per ;

    //Taking input from user
    cout << "Enter Base of Triangle " << endl ;
    cin >> base ;
    cout << "Enter perpendicular of Triangle " << endl ;
    cin >> per ;

    //Showing the Data user entered
    cout << "The Base and perpendicular are " << base
    << " , " << per << endl ;

    //Using Pythagoras Formula
    hyp = sqrt ( pow ( base , 2 ) + pow ( per , 2 ) ) ;
    cout << "The hypotenuse of Trianlge is " << hyp << endl ;
    return 0 ;
}
```

Output:

D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab Task 5\Task 5.1 A.exe

```
Enter Base of Triangle
9
Enter perpendicular of Triangle
3
The Base and perpendicular are 9 , 3
The hypotenuse of Trianlge is 9.48683

--------------------------------
Process exited after 3.126 seconds with return value 0
Press any key to continue . . .
```

Task 2

Write a program that will read in 3 grades from the keyboard and will print the average (to 2 decimal places) of those grades to the screen. It should include good prompts and labelled output. Use the examples from the earlier labs to help you. You will want to begin with a design. The Lesson Set 1 Pre-lab Reading Assignment gave an introduction for a design similar to this problem. Notice in the sample run that the answer is stored in fixed point notation with two decimal points of precision.

Code:

Task 5.1 A.cpp | Lab 5A-Task 1.cpp

```cpp
#include <iostream>
#include <iomanip>
using namespace std ;
int main ()
{
    //declaration of variables
    float g1, g2, g3, avg ;

    //Taking grades from user
    cout << "Enter first Grade " << endl ;
    cin >> g1 ;
    cout << "Enter second Grade " << endl ;
    cin >> g2 ;
    cout << "Enter third Grade " << endl ;
    cin >> g3 ;

    //Calculating the average
    avg = ( g1 + g2 + g3 ) / 3 ;

    //Setting precision for upto  two decimal places
    cout << fixed << setprecision (2) ;

    //showing output
    cout << "The Average of Grades is " << avg << endl ;

    return 0 ;
}
```

Output:

D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab Task 5\Lab 5A-Task 1.exe

```
Enter first Grade
97
Enter second Grade
98.3
Enter third Grade
95
The Average of Grades is 96.77

--------------------------------
Process exited after 13.05 seconds with return value 0
Press any key to continue . . .
```

Task 3

The Woody furniture company sells the following three styles of chairs:

Price per Chair

American Colonial = $85.00

Modern = $ 57.50

French Classical = $127.75

Write a program that will input the amount of chairs sold for each style. It will print the total dollar sales of each style as well as the total sales of all chairs in fixed point notation with two decimal places.

Code:

```cpp
#include <iostream>
#include <iomanip>
using namespace std;
int main ()
{
    // Declaring prices of Chairs
    float Achair = 85.00, Mchair = 57.50, Fchair = 127.75 ;

    // Declaring variabes
    int nofAC, nofMC, nofFC ;

    // Taking number of sales from user
    cout << "Enter number of American Chairs sold " << endl ;
    cin >> nofAC ;
    cout << "Enter number of Modern Chairs sold " << endl ;
    cin >> nofMC ;
    cout << "Enter number of French Chairs sold " << endl ;
    cin >> nofFC ;

    // Prinitng the sale of each Chair type seperately
    float saleAC, saleFC, saleMC, totalsale ;
    cout << fixed << setprecision (2) ;
    saleAC = Achair * nofAC ;
    cout << endl << "Total sales of American Chairs are " << saleAC << endl ;
    saleMC = Mchair * nofMC ;
    cout << "Total sale of Modern Chairs are " << saleMC << endl ;
    saleFC = Fchair * nofFC ;
    cout << "Total sales of French Chairs are " << saleFC << endl ;

    // Printing the Total sale of All Chairs
    totalsale = saleAC + saleMC + saleFC ;
    cout << endl << "The Total sale of all chairs is " << totalsale << endl ;

    return 0 ;
}
```

Output:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab Task 5\Lab 5A- Task 2.exe
Enter number of American Chairs sold
20
Enter number of Modern Chairs sold
15
Enter number of French Chairs sold
5

Total sales of American Chairs are 1700.00
Total sale of Modern Chairs are 862.50
Total sales of French Chairs are 638.75

The Total sale of all chairs is 3201.25

--------------------------------
Process exited after 6.04 seconds with return value 0
Press any key to continue . . .
```

Task 4

Write a program that will input total sales (sales plus tax) that a business generates for a particular month. The program will also input the state and local sales tax percentage. It will output the total sales plus the state tax and local tax to be paid. The output should be in fixed notation with 2 decimal places.

Code:

[*] Task 5.1 A.cpp | Lab 5A-Task 1.cpp | Lab 5A- Task 2.cpp | Lab 5A-Task 3

```
#include <iostream>
#include <iomanip>
using namespace std;
int main ()
{
    //Declaring variables
    float Tsales, STtax, LCtax, StateTax, LocalTax ;

    // Taking total sales and tax percentage from user
    cout << "Enter Total Sale of Month " << endl ;
    cin >> Tsales ;
    cout << "Enter Percentage of State Tax in Fractions " << endl ;
    cin >> STtax ;
    cout << "Enter Percentage of Local Tax in Fractions " << endl ;
    cin >> LCtax ;

    cout << fixed << setprecision (2) ;
    cout << endl << "The Monthly Sale is " << Tsales << endl ;

    // Calculating the State Tax
    StateTax = STtax * Tsales ;
    cout << "The State Tax for Monthly Sale is " << StateTax << endl ;

    // Calculating the Local Tax
    LocalTax = LCtax * Tsales ;
    cout << "The Local Tax for Monthly sale is " << LocalTax << endl ;

    return 0 ;
}
```

Output:

D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab Task 5\Lab 5A-Task 3.exe

```
Enter Total Sale of Month
1080
Enter Percentage of State Tax in Fractions
.06
Enter Percentage of Local Tax in Fractions
.02

The Monthly Sale is 1080.00
The State Tax for Monthly Sale is 64.80
The Local Tax for Monthly sale is 21.60

--------------------------------
Process exited after 10.46 seconds with return value 0
Press any key to continue . . .
```

Task 5

Modify the batavg.cpp by program until you get the correct result.

Code:

```
#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
    const int AT_BAT = 421;
    const int HITS = 123;
    float batAvg;

    // Cast HITS to float to ensure floating-point division
    batAvg = ( float ) HITS / AT_BAT ; // Corrected assignment statement
    cout << "The batting average is " << batAvg << endl; // Output the result

    return 0;
}
```

Output:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab Task 5\Type casting.exe
The batting average is 0.292162

--------------------------------
Process exited after 0.1117 seconds with return value 0
Press any key to continue . . .
```

Thank You for Your Time!

<u>THE END</u>